

Improving English-Bangla Translation in Low-Resource Settings Using Back-Translation

Master's research project report asset

This PDF summarizes the measured final experiment included in this repository. It is intended as a report-ready results and discussion artifact, not as a replacement for university formatting or supervisor review.

Research Objective

Research question: Does back-translation improve English-Bangla translation performance in a low-resource setting?

The project compares a baseline google/mt5-small English-to-Bangla model against an improved model trained with original data plus filtered synthetic back-translated data.

Dataset and Configuration

Dataset: ai4bharat/samanantar, configuration bn.
Full Samanantar Bangla training split metadata: 8,604,580 sentence pairs.
Final low-resource setting: 5,000 training pairs, 500 validation pairs, 500 test pairs.
Synthetic data: 3,000 raw candidates generated; 2,000 filtered synthetic pairs retained.
Hardware: Tesla T4. Optimizer: Adafactor. Learning rate: 0.0001. Effective batch size: 16. Epochs: 3 baseline, 3 reverse, 3 improved.

Measured Results

Baseline mT5-small BLEU: 0.2359.
Improved filtered BT mT5-small BLEU: 0.2114.
BLEU change: -0.0244 (-10.36%).

Baseline validation loss: 3.1324.
Improved validation loss: 3.0680.
Validation-loss change: -0.0644.

Training times: baseline 735.46 seconds, reverse model 609.03 seconds, improved model 940.81 seconds.

Conclusion

The stronger filtered back-translation setup did not improve BLEU for English-Bangla translation in this low-resource experiment. Validation loss improved, but BLEU decreased. The most likely explanation is that the reverse model generated synthetic English that passed surface-level filters but still introduced semantic noise.

Limitations and Future Work

Limitations: small low-resource subset, noisy reverse-model outputs, limited synthetic data quality, T4 runtime/memory constraints, and BLEU sensitivity to lexical overlap.

Future work: train on a larger subset, use stronger reverse models, improve semantic filtering, increase synthetic data quality, train for more epochs, and compare mT5-small with alternatives such as mBART or larger mT5 variants.